9 Experiments

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Experiments, alone or in combination with other methods of data gathering, are growing in popularity among sociolinguists because they can provide different kinds of data that contribute to our understanding of social variation in language use. Carefully designed production experiments allow us to efficiently collect large quantities of speech that directly bear on our research questions. Complementary data from perception experiments provide concrete evidence for how social variation in speech is perceived and interpreted by non-linguists. Although many experiments rely on highly constrained forms of speech, such as read words or sentences, more natural production and perception data can be obtained through the use of interactive tasks that both constrain linguistic content and allow participants to converse more naturally.

Production Experiments in Sociolinguistics

A primary strength of production experiments is their efficiency. In an experiment, the target list of utterances (words, sentences, or constructions) is designed to elicit an answer to the research question and is established by the experimenter before any data are collected. As a result, we can ensure that the phenomenon that we are interested in will be elicited a sufficient number of times from each participant over the course of the experiment. In ethnographic observation or sociolinguistic interviews, by contrast, we may need to record many hours of speech from a single participant before the phenomenon of interest is produced a sufficient number of times for analysis.

This experimental strength is particularly notable when we are studying a relatively rare phenomenon, such as the vowel /oj/ or modal constructions. For example, in two randomly selected interviews from the Buckeye Speech Corpus (Pitt et al., 2007), the vowel /oj/ occurred only 12 times in 43 minutes in one interview and only six times in 69 minutes in the other. Similarly, modals (including *can*, *might*, *would*) occurred only 43 times in the first interview and 46 times in the second interview. Whereas these sociolinguistic interviews did not elicit many examples of these phenomena, we can specifically target the linguistic categories that we are interested in when we design an experiment. If we wanted to explore the monophthongization of /oj/ in regional varieties of American English, for example, we could supplement a sociolinguistic interview, in which we might elicit only a small number of /oj/ tokens, with a short word list reading task to ensure that we have enough examples of /oj/ to confidently answer our question. Similarly, if we wanted to study the double modal construction (e.g., *might could*) in American English, we could design a story completion task to elicit a sufficiently large sample of modal constructions from each of our participants.

This data-gathering efficiency applies to virtually all well-designed production experiments and allows us to explore variation in linguistic categories at many levels of linguistic structure, including segments, prosody, and morphosyntax. Segmental variation can be explored through word list, sentence list, and paragraph reading tasks, in which participants are recorded reading aloud a carefully constructed set of materials that manipulates the variables of interest to the researcher. For example, regional vowel variation has been examined in American English (Clopper, Pisoni, & de Jong, 2005), Dutch (Adank, van Hout, & van de Velde, 2007), and Portuguese (Escudero, Boersma, Rauber, & Bion, 2009) using word list reading tasks, either with the words embedded in a simple carrier sentence (Dutch, Portuguese) or not (English). Prosodic variation can also be examined using reading tasks, although longer passages and scripted dialogues are often used to ensure that the intended information structure is conveyed to the participants. For example, Clopper and Smiljanic (2011) examined regional prosodic variation in American English using read paragraphs, and Arvaniti and Garding (2007) and Elordieta and Calleja (2005) used scripted dialogues between the experimenter and each participant to explore prosodic variation in American English and Spanish, respectively. More interactive tasks, such as the map task (Anderson et al., 1991), in which two participants work together to negotiate a route along a map, can also be used to elicit prosodic variation (e.g., Barry, 2007). Morphosyntactic variation cannot easily be examined using reading tasks, because participants' ability to read a given construction is independent of the extent to which they use that construction in non-scripted social interactions. However, with a little creativity on the part of the researcher, partially scripted tasks (e.g., sentence completion, story completion, and responses to questions by the experimenter) and interactive tasks (e.g., the map task) have the potential to provide us with multiple examples of the variable(s) of interest from each participant. For example, Balcetis and Dale (2005) used a picture description task to explore syntactic priming as a function of the social relationship between the participants. Participants were more likely to describe pictures using the same syntactic structure (e.g., passive vs. active voice) as an unrelated prime sentence produced by the experimenter if the experimenter was perceived as "nice" rather than "mean."

A second primary strength of production experiments is the apples-to-apples comparisons that they permit. Many factors contribute to the phonetic, prosodic, and morphosyntactic realization of an utterance, and experiments allow us to control the factors that we are not interested in so that we can focus on the factors that we are interested in. The effects of preceding and following consonantal context on vowel variation have long been recognized in the sociolinguistic community (e.g., Labov, 1972). More recent research has demonstrated that the vowel in the following syllable can also affect the realization of a target vowel (Cole, Linebaugh, Munson, & McMurray, 2010), suggesting that coarticulatory effects extend over multiple segments. Similarly, morphological context is a well-known factor affecting variable processes such as consonant cluster reduction in African American English (Guy, 1980). Finally, semantic and discourse contexts also affect the realization of individual segments. Words in semantically predictable contexts tend to be reduced relative to words in less predictable semantic contexts (Lieberman, 1963), and words that are repeated in a discourse tend to be reduced relative to the first time they are produced in the discourse (Fowler & Housum, 1987). These discourse-level effects of predictability and repetition also interact with prosodic structure. Words referring to new referents in a discourse tend to be prosodically prominent, and prosodically prominent words are hyper-articulated relative to less prominent words (de Jong, 1995). Segmental strengthening is also observed at prosodic boundaries (Dilley, Shattuck-Hufnagel, & Ostendorf, 1996). Thus, contextual effects emerge not just from immediately neighboring segments but also from more distant segments and higher level linguistic structure, including prosody, morphology, and semantics.

Independent of the discourse context, properties of words themselves also have a substantial effect on their phonetic and syntactic realization. For example, high-frequency words tend to be reduced relative to low-frequency words (Munson & Solomon, 2004), words that are phonologically similar to many other words are hyperarticulated relative to words that are phonologically similar to few other words (Wright, 2004), and content words are more likely to be prosodically prominent than function words (Calhoun, 2010). Similarly, some variation in syntactic constructions, such as the dative alternation in English, can be attributed to individual word biases. For example, whereas the recipient of *bring* is very likely to have been previously mentioned in the discourse context, the recipient of *take* is much more likely to be new in the discourse context (Bresnan, Cueni, Nikitina, & Baayen, 2007). As a result, *take* is more likely to be produced in a dative PP construction (e.g., *take the bag to school*) than *bring*, which is more likely to be produced in a dative NP construction (e.g., *bring me the book*). Finally, the use of specific words or phrases also contributes to syntactic variation. For example, polarity and the selection of a pronoun vs. a full noun phrase affect the realization of *was/were* variation in British English (Cheshire & Fox, 2009; Tagliamonte, 1998). Thus, the specific words that a talker uses to convey his or her message will also have a substantial impact on how the variables of interest are realized.

In an experiment, these contextual and lexical effects can be controlled within and across participants, which increases the likelihood that true effects will be observed and decreases the likelihood that spurious results will be interpreted as significant. For example, in an experiment exploring degrees of /u/ fronting by male and female speakers of Southern American English, each participant could be recorded producing the same set of 20 target words in isolation or embedded in a carrier phrase or paragraph. By controlling for consonantal context, lexical variability, and discourse context, we would be more likely to observe a significant difference between genders if one actually existed. If instead we were to extract words containing /u/ from interviews with the same set of participants, we would not be able to control as well for phonological, lexical, and discourse properties in our selection of target words, and the true effects of gender could be obscured. For example, if the tokens produced by the female talkers happened to be in low-frequency words, and the tokens produced by the male talkers happened to be in high-frequency words, the /u/ productions might appear to be equally fronted across genders. That is, a true gender effect could be masked by the frequency effect: the male and female /u/s would appear to be overlapping because the female talkers produced relatively more peripheral (i.e., backed) /u/s in their low-frequency words, and the male talkers produced relatively less peripheral (i.e., fronted) /u/s in their high-frequency words. In addition, a spurious gender effect might be obtained for /u/ raising: the female /u/s would appear to be higher than the male /u/s because the female talkers produced relatively more peripheral (i.e., raised) /u/s in their low-frequency words, and the male talkers produced relatively less peripheral (i.e., lowered) /u/s in their high-frequency words.

In spontaneous and interview speech, the researcher typically has very little control over the many factors that contribute to variation in speech production, and the resulting data are often very noisy. Although our statistical models are improving to allow us to include many different contributing factors in our analyses of production data, we must still identify all of the potentially relevant factors, quantify those factors in an appropriate and meaningful way, and avoid overfitting our data by including too many variables in our analysis. In addition, noisy datasets typically require more data points than clean datasets to observe true results and to avoid spurious results. In production experiments, the researcher has much more control over the materials, and carefully designed experiments can yield valid and highly reliable data. Having comparable data from each participant allows us to be more confident that any differences that we observe across participant groups are due to social factors rather than accidental linguistic factors.

Perception Experiments in Sociolinguistics

The primary strength of perception experiments is that they allow us to explore how the variation that we observe in production is used by non-linguists to interpret the intended message and to identify social characteristics of the talker. These kinds of perceptual judgments are essential for a complete understanding of sociolinguistic variation. One classic example of the important contribution of perceptual judgments to sociolinguistic research is the phenomenon of near-mergers. In a near-merger, language users maintain a phonetic distinction between two phonemes in production but report that minimal pairs containing those two phonemes are the same in a perception task (Labov, Karen, & Miller, 1991). Thus, a near-merger is a perceptual phenomenon that we cannot observe directly from production data.

Direct linguistic judgment tasks, such as grammaticality judgment tasks or the minimal pair test used by Labov et al. (1991), tap participants' explicit knowledge of linguistic structure and allow for conscious reflection by the participants. When combined with an interview in which the participants are encouraged to explain or discuss their linguistic judgments, these tasks can provide very rich data on segmental and morphosyntactic variation. However, because direct tasks may allow for conscious reflection by the participants, the data may be colored by the participants' prescriptive notions of grammaticality or knowledge of orthography. Indirect tasks that focus on processing or interpreting the linguistic content of an utterance may therefore be preferable for exploring what participants actually do, rather than what they believe to be true about their language. For example, tasks that require participants to respond as quickly as possible, such as lexical decision tasks (Floccia, Girard, Goslin, & Konopczynski, 2006) or speeded classification tasks (Clopper & Pate, 2008), can be used to examine the processing difficulties associated with an unfamiliar variant. Slower response times are associated with difficult processing, whereas faster response times are associated with easy processing. Similarly, familiarity or exposure to a particular variant can be examined using semantic priming (Sumner & Samuel, 2009; Warren & Hay, 2006). Familiar phonological variants will effectively prime semantic associates (e.g., *chair* primes *sit*), but less familiar variants will not. This priming effect is realized in perception tasks as faster response times to primed targets than to unprimed targets. These tasks that involve response time data are very sensitive to subtle differences in linguistic processing and therefore have the potential to uncover aspects of sociolinguistic variation that are more difficult to observe in production experiments, such as passive competence in a second dialect, or in linguistic judgment tasks, such as familiarity with a particular variant.

Perception experiments can also be used to determine how the social categories that we identify in our production data map onto the social categories of non-linguists. Although production data inform us about the variation that exists in the world, the researcher is ultimately responsible for carving up the data into relevant social categories for interpretation. Perception experiments can provide essential complementary evidence for the relevance of the social categories for the local community. For example, if neighborhoods within an urban area partially overlap with social class divisions, a perception experiment could help determine which of the two variables is more central to social identity in the local community. We can imagine a study in which participants are asked to categorize talkers with varying social and geographic backgrounds into groups first based on neighborhood and then based on social class. If neighborhood is a more important category for the participants than social class, performance across participants should be more consistent in the neighborhood classification task than in the social class task. Alternatively, if social class is more central for the participants than neighborhood, greater consistency should be observed in the social class task than the neighborhood task. This experimental design would also have the potential to capture the relevant neighborhood or social class distinctions for the participants. If the neighborhoods identified by the researcher were either too broadly or too narrowly defined, the mismatch between the researcher's categories and the participants' categories would be revealed by the pattern of responses.

The perceptual dialectology tasks that Preston (1989) has developed, including map drawing, correctness ratings, and pleasantness ratings, are all motivated by this question of the relationship between sociolinguists' maps of linguistic variation and the cognitive maps of non-linguists. Perceptual dialectology is primarily concerned with participants' beliefs about linguistic variation and therefore does not typically involve explicit perception of a particular stimulus. However, perception experiments can also be used to obtain perceptual judgments in response to variable linguistic stimulus materials. For example, Clopper and Pisoni (2004, 2007) examined how participants identify the regional dialect of unfamiliar talkers using forced-choice categorization tasks (in which listeners assign each talker to one of a researcher-defined set of categories) and free classification tasks (in which listeners group talkers together without any predefined categories). Finally, perception experiments can be used to elicit attitude judgments about talkers based on their speech (see Campbell-Kibler, Vignette 8a). One of the most prevalent paradigms in attitude research is the matched-guise technique (Lambert, Hodgson, Gardner, & Fillenbaum, 1960), which has been used to explore attitudes toward both phonological variation, such as regional and foreign accents (e.g., Giles, 1970), and grammatical variation, such as copula absence in African American English (e.g., Bender, 2005).

Perception experiments also provide opportunities to explore the relationships between linguistic and social categories in speech processing. For example, Clopper and Pisoni (2004) used regression techniques to determine the phonetic properties that contribute to the identification of the regional dialects of unfamiliar talkers, and Strand (1999) observed differences in /s-ʃ/ identification as a function of the gender and gender typicality of the voices. Thus, perception experiments can be used to examine how linguistic variation affects social categorization as well as how social variation affects linguistic categorization.

Potential Limitations of Experiments in Sociolinguistics

The primary limitation that is often identified for experiments for sociolinguistic research is the lack of naturalness that these tasks may involve. To capitalize on the strengths of experimental design and maximize the control over the materials, read speech is often elicited in production experiments and presented to listeners in perception experiments. However, read speech is known to differ from spontaneous speech at all levels of linguistic structure and is therefore not representative of all kinds of speech. In addition, the use of read speech in production experiments requires participants to be literate, which may limit the population of potential participants in an undesirable way. Production experiments involving read speech are also infeasible for varieties that do not have a relatively standardized written form. The problems with read speech have received particular attention in the prosody literature, and many researchers are developing novel interactive tasks that constrain linguistic content but allow for the collection of relatively natural speech. These tasks include the map task mentioned earlier (Anderson et al., 1991), as well as the diapix picture description task (Van Engen et al., 2010), partially scripted games (Speer, Warren, & Schafer, 2011), and the holiday tree decorating task (Ito & Speer, 2006). Each of these tasks was developed with a somewhat different research goal and is therefore structured somewhat differently. However, these tasks have several properties in common that are central to developing any kind of successful interactive task: two participants interact with each other to achieve a shared goal, and the materials are carefully designed to elicit the target utterances. These approaches have the potential to provide us with controlled production materials in spontaneous, interactive speech.

A second potential limitation of experiments as a method of data gathering in sociolinguistics is that phonetics and psycholinguistics experiments are traditionally run with university students using high-quality sound equipment in quiet locations. However, university students are not representative of the adult population, particularly with respect to literacy, computer skills, and experience with formal testing. Thus, experimental paradigms that are effective with university student populations may not work with other populations that are of interest to sociolinguists. In addition, bringing participants into a laboratory setting at a university is a particular social context that may have substantial effects on how participants perform. Now that high-quality microphones and digital recording equipment are more portable and affordable than ever, it is becoming possible to conduct both production and perception experiments in the field. Another option is a web-based perception experiment, such as Campbell-Kibler's (2007) matched-guise study of the (ING) variable in American English. However, the quality of the audio output that individual participants experience is difficult to control in these studies. Thus, web-based experiments are often more suitable for studies of lexical or syntactic variation, in which fine phonetic detail is not central to the research question, and for experiments involving perceptual dialectology tasks or survey methods that do not require perceptual responses to auditory stimulus materials.

Methodological Considerations in Experimental Design

Regardless of the experimental paradigm that we adopt, we need to carefully select our stimulus materials. In particular, we need to ensure that we have the right kind of linguistic materials (e.g., words, sentences, passages, or dialogues) to address our research question and that we have controlled for as many as possible of the potentially relevant factors (e.g., segmental and semantic contexts, frequency, etc.) that may affect the variable(s) of interest. Longer stimulus materials will necessarily require greater care in controlling for these other factors, but there are many ways to control for these sources of variability without creating an unmanageably large experiment. Variables that we are interested in should be balanced (i.e., appear equally often) across experimental conditions, whereas variables that are not central to our research question can be controlled by selecting a single level of the variable for all of our materials (e.g., choosing only high-frequency words) or allowing the materials to vary freely with respect to that variable across experimental conditions (e.g., choosing words of varying frequencies for all conditions).

In addition, we typically do not want participants to know (or be able to figure out) our precise research question, because their behavior may change if they are trying to help us get good data. For example, if we are interested in a potential vowel merger, we may not want to ask our participants to read a list of minimal pairs containing the relevant contrast, because they may try to produce the targets differently simply because they know they are supposed to be different words. Similarly, if we are interested in the prosody of compound nouns, we may not want to label all of the landmarks in a map task with compound nouns, because participants may notice the pattern and produce particularly marked prosody on the targets. One effective way of reducing the possibility that participants will figure out what the experiment is about is to use fillers. Fillers are extra stimulus materials that are unrelated to the target materials and serve to maintain some apparent randomness in the complete set of materials that the participant is exposed to. In our experiment about vowel mergers, the fillers could be words containing vowels that are not involved in the merger. In our experiment about compound nouns, the fillers could be proper nouns or adjective–noun sequences.

Even if participants are not aware of the purpose of the experiment, their experience with some of the stimulus materials may affect how they perform on other materials. For example, if the minimal pairs in our vowel merger experiment are presented one after the other, participants may produce a larger difference between the pairs than if they are separated by several fillers. Presenting materials in a random order can reduce these effects, known as order effects. Randomization can either be performed once, so that each participant is exposed to the materials in the same random order, or separately for each participant. Different randomizations for each participant allow the effects of order to vary randomly across items and participants, and are effective for larger experiments with many participants. For smaller experiments with fewer participants, a single random order may be more appropriate so that order effects are constant across participants and can be included as a factor in the statistical analysis. A second method for controlling for order effects is counterbalancing, in which the order of experimental conditions is balanced across participants. For example, in our compound noun experiment we might have one map with a "white house" landmark and one map with a "White House" landmark so that we can compare the prosody of the adjective–noun sequence ("white house") to the prosody of the compound noun ("White House"). To counterbalance for order effects, half of our participants should complete the map with "white house" first, and the other half of our participants should complete the map with "White House" first, so that any effects of having already encountered the other form will be balanced across the two participant groups.

Finally, the "observer's paradox" is a well-known problem for many kinds of data-gathering methods in sociolinguistics (Labov, 1972), including experiments. Accommodation in speech production is well-documented among interlocutors (e.g., Giles, 1973) and has been observed in sociolinguistic interviews (e.g., Rickford & McNair-Knox, 1994) and interactive laboratory tasks (e.g., Pardo, 2006). Hay, Drager, and Warren (2010) have also found effects of experimenter dialect on performance in a speech perception task that was independent of the variation in the stimulus materials. Similarly, Goldinger and Azuma (2003) found that the experimenters' expectations about the outcome of the experiment biased participants to perform in a particular way. Thus, the experimenter plays a crucial role in how participants perform, and it is important to ensure that all experimenters are well trained. If more than one experimenter is involved in a particular experiment, randomization and counterbalancing techniques should be used to ensure that any effects of experimenter bias or social characteristics are evenly distributed across the data.

Conclusion

Experiments allow us to efficiently obtain large amounts of relevant data. Although many experimental paradigms involve less natural speech than other methods of data gathering in sociolinguistics, more interactive tasks can be used to elicit comparable target utterances containing the variable(s) of interest from all participants. Thus, experiments have the potential to complement ethnographic, sociolinguistic interview, and survey methods to provide converging evidence for both descriptive observations and theoretical claims.

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